

ISOMETRIC DRAWING AUTO-EDIT

SP3D Isometric Auto-Edit

Batch-corrects SmartPlant 3D isometric drawings: insulation / dimension / weld-no / elbow / elevation text, item-no rotation and support symbols.

Key Purpose

Why this program

1. Batch-fix SP3D isometric drawings in one click

- Runs a 12-step pipeline (00~12) on every selected DXF at once, saving <name>_all.dxf
- **Effect:** Hours of manual redlining per drawing set are replaced by a single unattended pass

2. Standardize insulation, dims, welds, elbows & elevations

- Deletes insulation clutter, edits dimensions, weld-no, elbow and elevation text automatically
- **Effect:** Every isometric comes out to the same house standard — no reviewer-to-reviewer inconsistency

3. Auto item-no rotation/position & support symbols

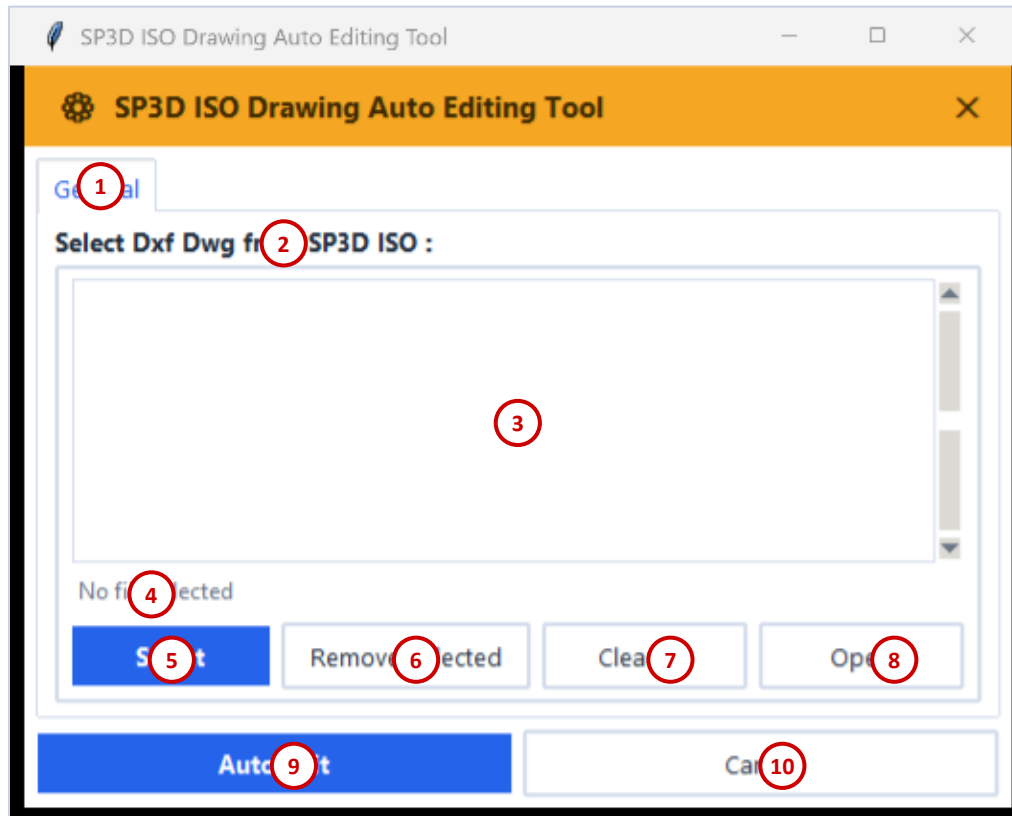
- Rotates and repositions item-number tags, trims strays, re-tags supports and support symbols
- **Effect:** Eliminates the tedious tag cleanup that is easy to miss and cause rework at issue

4. In-memory, offline, no CAD license required

- Loads each DXF once, applies all steps in memory, writes once — no SP3D or AutoCAD session
- **Effect:** Runs on any PC with Python; no CAD seat is tied up and no license cost per drawing

Required inputs : SP3D isometric DXF drawings

Main Window – Item Guide



1. General tab
2. "Select Dxf Dwg from SP3D ISO" label
3. Selected-DXF list box
4. File-count status ("No file selected")
5. Select (add DXF files)
6. Remove Selected
7. Clear All
8. Open (view highlighted DXF)
9. Auto Edit (run 12-step pipeline)
10. Cancel (close window)

Processing Steps

What 'Auto Edit' runs, in order

1. Polyline to line
2. Insulation delete
3. Dimension edit
4. Weld-no edit
5. Elbow edit
6. Elevation text edit
7. Item-no rotate
8. Item-no reposition
9. Trim delete
10. Instrument edit
11. Support tag
12. Final cleanup

Overview

What it does

1. Batch-corrects SmartPlant 3D isometric drawings: insulation
2. Dimension
3. Weld-no
4. Elbow
5. Elevation text
6. Item-no rotation and support symbols

Category: Isometric Drawing Auto-Edit

Folder: ISO_AUTO_EDIT/ISO_AUTO_EDIT_FOR_SP3D

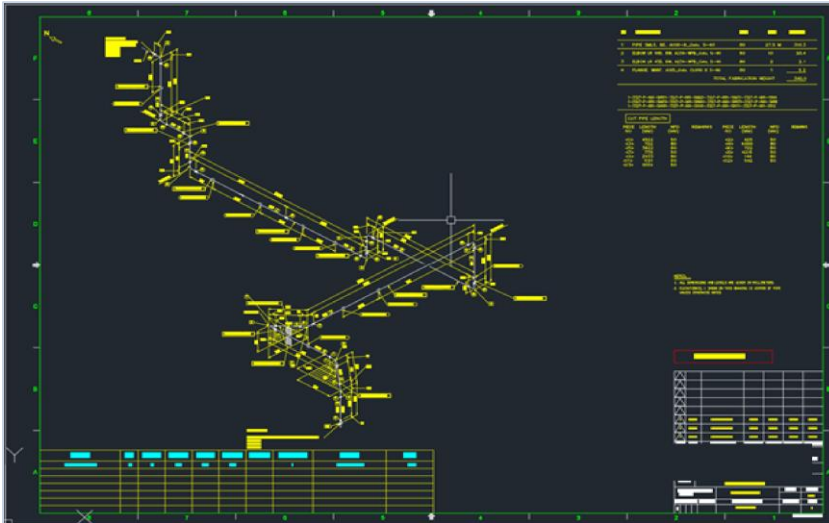
Main scripts: SP3D_ON_RUNTEST.PY

Scripts / Status: 33 scripts · Operational

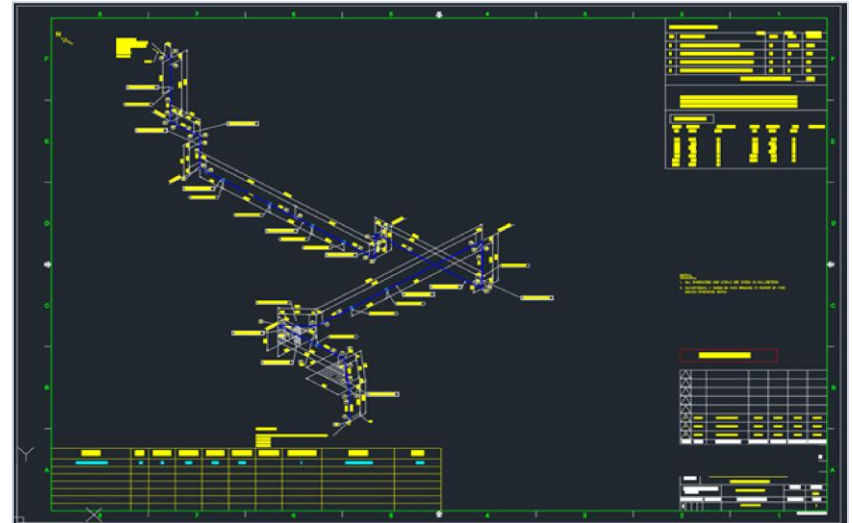
Tags: SP3D · DXF · BOMFORM · SUPP

Output Samples

Output Samples



SP3D ISO Input File



SP3D ISO Auto Output File

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Isometric Drawing Auto-Edit.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: SP3D_ON_RUNTEST.PY.
- Output samples and the program window above reflect the current build.

Thank You

SP3D Isometric Auto-Edit · Plant Engineering Total Service System